

Tidebid — Boat Auction Platform: Architecture & Build Spec (for Codex)

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Start here

We are building a timed online auction platform for boats in four languages, matching boatauction.com's proven model and beating it on live bidding, safe payment, transparency and seller acquisition. Codex builds it milestone by milestone from this document; you review and merge.

"Tidebid" is a placeholder brand. Replace it everywhere once you pick a name and domain.

The two prototype files, `tidebid_platform.tsx` and `vendre_mon_bateau_estimator.tsx`, are visual references only; keep them in `docs/prototypes/`. The first one was written before this spec and its bidding code differs from it. Whenever a prototype and this document disagree, this document wins.

How to run this with Codex

1. Create an empty GitHub repository and connect it to Codex.
2. Save the AGENTS.md section of this doc as `AGENTS.md` at the repository root. Export this whole doc to Markdown and save it as `docs/SPEC.md`.
3. Run the milestones in the Build plan section in order. One milestone = several tasks. One task = one pull request.
4. For every task, paste its prompt, ask Codex for a plan first, approve the plan, then let it implement.
5. Merge only when CI is green and the acceptance criteria listed with the task pass.
6. If Codex says the spec is unclear, answer by editing this doc, then re-export it. The doc stays the single source of truth.

Decisions already taken

Codex must not re-open these choices. Each one keeps the system simple, testable and EU-hosted.

Area	Choice	Why
Language	TypeScript (strict) everywhere	One language for web, API, workers and shared rules

Area	Choice	Why
Repository	pnpm workspaces + Turborepo monorepo	Shared domain code, one CI
Public site, seller portal	Next.js (App Router), server-rendered	SEO for every lot page, fast first load
Admin panel	Separate Next.js app on its own subdomain	Isolated access, stricter security
API, real-time, webhooks	NestJS on the Fastify adapter	Guards for permissions, built-in WebSocket and job modules
Database	PostgreSQL 17+ with Prisma	Transactions and row locks for bids
Jobs, cache, pub/sub	Redis + BullMQ	Lot closing timers, notifications, fan-out
Live updates	Socket.IO with the Redis adapter	Rooms per lot and per user, reconnects
Search	Meilisearch	Facets, typo tolerance, 4 languages
Files	S3-compatible storage + CDN	Photos, videos, documents
Hosting	Docker containers in an EU region (default AWS eu-west-3, Paris), Terraform	Data stays in the EU
Locales	fr (default), en, nl, de	France first, Benelux and Germany next

The payment, e-signature and identity-check vendors are chosen in the Payments and Compliance sections. They sit behind adapters, so a vendor can change without touching business logic.

The three rules that make the platform trustworthy

1. The server decides everything. Price, leader, time left and closing are computed only on the server, inside one database transaction per bid.
2. Every rule lives once, in a shared package of pure, tested functions: bidding, fees, deadlines and the seller estimator.
3. Nothing important happens silently. Every change to a lot, bid, payment or payout writes an audit entry, and the admin panel can see and control all of it.

How online auction platforms work

Boat auctions online are timed ascending auctions. Each lot has its own closing time, bids only go up, a private maximum bid bids for you, and a late bid extends the lot so nobody wins by sniping. After closing, the seller confirms the sale (or, in a regulated auction, reaching the reserve confirms it automatically) and the money flows through a trusted third party until handover.

Vocabulary, with the default rule we adopt

Term	What it means	Our default
Auction (sale)	A dated event that groups many lots	One main auction per month, viewing day the Saturday before
Lot	One boat (or engine, trailer) for sale	Own page, own closing time
Start price	Where bidding opens	Set with the seller
Increment	Smallest step above the current bid	Table by price band, admin-editable
Single bid	One exact amount, placed once	Whole euros, at least the next minimum
Max bid (automatic bid)	A private ceiling; the system bids the smallest step needed for you	Private to the bidder and admins
Reserve price	Hidden minimum the seller accepts	Optional; in regulated mode reaching it awards the lot, in brokerage mode it is a target for the seller
No reserve	Highest bid wins, whatever the amount	Badge on the lot; regulated mode only
Soft close	A late bid pushes the closing time back	Bid in the last 5 min → closes 5 min after that bid
Staggered closing	Lots close one after another	1 minute apart
Hammer price	The winning bid	Basis for every fee
Buyer's premium	Platform fee paid by the buyer on top of the hammer price	18% / 12% / 8% by start price, plus VAT on the premium
Subject to award	Seller may still accept or decline the top bid	Seller has 72 hours

Term	What it means	Our default
Second-chance offer	Lot offered to the runner-up when the winner defaults	Runner-up's last bid
Escrow	Buyer's money held by a licensed third party	Released to the seller after handover
Viewing day	Buyers inspect the boat in person	Saturday 10:00–12:00

What the leading platforms do

Platform	Late-bid extension	How lots close	Max bids and reserve	Non-payment protection
boatauction.com	Bid in last 5 min adds 5 min (source)	Monthly sale, lots about 1 min apart	Normal or automatic bid; hidden minimum; seller decides within 72 h (terms)	Premium + VAT stays due as a break-up fee
Catawiki	Bid in last 60 s adds 90 s	Staggered, about 1 min apart	Max above reserve jumps to the reserve; max below reserve is bid in full; equal maxima: earlier one leads	Card hold on risky bids, at most €1,000 in total
Troostwijk	Bid in last 2 min adds 2 min	Lot by lot	Auto-bid	Public auction: bidders may attend the closing in person, so no cooling-off period
iRostrum (UK)	Bid in last 5 min extends up to 5 min, repeatedly	About 2 min apart; one lot's extension never moves another	Raising your own max never bids against yourself	—

We copy the strongest rule from each: 5-minute soft close with independent lots, Catawiki's reserve-aware max bids and tie rule, and card holds for risky bidders.

What boatauction.com does today

The competitor is VesselAuction B.V. (Naarden, NL), also trading as bootveiling.com, vesselauction.com and superyachtauction.com ([terms](#)).

- **Calendar.** One auction a month, closing on a Monday evening, viewing day the Saturday before, 10:00–12:00 ([calendar](#)).
- **Buyer fees.** 18% premium up to €25,000, 12% from €25,000 to €100,000, 8% from €100,000, chosen by start price, plus 21% VAT on the premium ([source](#)).
- **Seller fees.** Flat by length: €100 up to 5 m, €200 for 5–10 m, €300 for 10–15 m, €500 above 15 m ([source](#)).
- **After closing.** Seller has 72 hours to accept. Buyer pays a third-party funds account within 4 days of the invoice, collects within 14 days and becomes owner after payment and signing, or 14 days after award at the latest.
- **Trust.** A bailiff supervises the auctions. Bidders may attend the closing at the Naarden office, which makes it a public auction where the statutory right of withdrawal does not apply ([terms](#)).
- **Seller lock-in.** Exclusivity until closing plus 30 days; withdrawing costs the premium plus VAT on the current bid, at least €500.

Where we beat them

- **Live bidding.** Prices, leader and clock update instantly; their page needs a reload.
- **Smarter bids.** Max bids that meet the reserve, and, in regulated mode, automatic award when the reserve is reached, so sellers wait less.
- **One set of deadlines.** Their own pages contradict each other on payment (2 or 4 days) and seller payout (24 hours or 5 working days); ours come from one config.
- **Digital closing.** Instant invoice, licensed escrow, e-signed transfer document and a handover code instead of paper and email.
- **Seller acquisition.** A "Vendre mon bateau" funnel with a cost-of-waiting estimator, a live seller dashboard and a public results archive.
- **Search visibility.** Catalogue pages rendered on the server, so search engines see every lot.

Product scope

Version 1 ships three products on one codebase: the public auction site, the seller portal and the admin panel. Two journeys must work end to end before launch, and each becomes an automated end-to-end test.

1. **Buyer:** discover → register → verify → bid → get outbid → set a max bid → win → pay into escrow → e-sign the transfer → collect with the handover code.
2. **Seller:** estimator → lead → intake wizard → verification → e-signed consignment → listed → live → accept the top bid → handover → payout.

Roles

Role	Who	Can do
Visitor	Anyone	Browse, search, use the estimator, read guides, watch lots on this device
Bidder	Registered, email and phone verified	Bid within their credit limit; ID check required for any bid of €25,000 or more
Private seller	Owner of a boat	Consign boats, follow stats, accept or decline, get paid
Business seller	Dealer, broker, bank, leasing firm, bailiff	Everything a seller does, plus bulk listing and VAT invoices
Staff	Platform employees, several admin roles	Defined in the Admin panel section
Supervisor	Bailiff or notary	Read-only access to bid logs and closing reports

Buyer features

Feature	Done when
Catalogue and search	Filters for type, price, length, year, country, brand, fuel, engine hours, no reserve and closing within 24 h; facet counts; sort by closing soonest, price, newest; filters live in the URL; first page rendered on the server

Feature	Done when
Lot page	Up to 80 photos, video, documents; key facts; specs grouped by the category template; approximate map (about 2 km); viewing-day sign-up; question form; similar lots; structured data for search engines
Bidding panel	Price, leader and clock update within 1 s; single and max bids; quick-bid amounts; cost breakdown before the binding confirm; reserve badge when the lot shows it; a clear message for every rejection reason
Watchlist and saved searches	Heart on every card; reminders 24 h, 1 h and 10 min before closing; alerts when new lots match a saved search
Notifications	In-app, email and web push; optional SMS when outbid in the last hour; preferences per type
My bids	Tabs for leading, outbid, won, lost and awaiting seller; raise or cancel a max bid; invoices and documents
After winning	Invoice within 5 minutes; pay by bank transfer to a unique IBAN or a local method; e-sign the transfer; book a pickup slot; show a handover code; confirm receipt
Results archive	Every closed lot with its outcome and, when the seller allows, the final bid

Seller features

Feature	Done when
"Vendre mon bateau" button	In the header on every page, sticky on mobile, in the footer and on lot pages ("Own a similar boat?")
Cost-of-waiting estimator	Specified in the Seller acquisition section
Intake wizard	Five steps: owner and contact, boat basics, equipment and condition (per category), photos and documents, auction date and viewing availability; autosave on every field; resume by magic link; camera upload on mobile

Feature	Done when
Verification	Private: ID, address, IBAN, tax number. Business: company registration, VAT number checked in VIES, beneficial owners
Consignment agreement	Generated from a template with the boat, fees and exclusivity terms; e-signed; signed PDF stored
Listing build	AI drafts the description and translations; staff edits and approves before publishing
Seller dashboard	Views, watchers, viewing sign-ups, questions and live bids; weekly email report
Award decision	Top bid, runner-up and reserve status; accept, decline or counter-offer; 72 h countdown; reminders at 24 h, 48 h and 66 h
Payout	Released 24 h after handover is confirmed, or after 14 days when a consumer buys from a business outside public-auction mode; payout statement PDF; status visible

Business sellers and partners

Banks, leasing firms and bailiffs sell repossessed boats, as boatauction.com does with its foreclosure auctions. They get bulk upload by CSV, special conditions per lot and their own VAT invoices.

Later, not in version 1: native mobile apps (the site is an installable web app first), live-streamed auctioneer sales, sealed-bid tenders, Buy Now after an unsold auction, transport and survey partners, and currencies other than EUR.

Auction rules and bidding engine

One pure function decides every bid: `applyBid(lotState, command, now, rules)`. It lives in `packages/domain`, has no database or clock access, and is replayable. The API wraps it in a single locked transaction; nothing else may change a lot's price, leader or closing time.

Rules and their defaults

All values live in a versioned settings table that admins edit. Each lot stores the settings version it went live with.

Setting	Default	Notes
Bid unit	Whole euros	Stored as integer cents
Soft-close window	5 min	Only bids that change the price or the leader count
Extension	Closes 5 min after that bid	$\text{endsAt} = \max(\text{endsAt}, \text{bidTime} + 5 \text{ min})$, no cap
Closing interval	60 s between lots	Each lot extends on its own
Reserve-aware max bids	On	A max at or above the reserve jumps to the reserve; below it, the max is bid in full
Auto-award at reserve	By sale mode	Regulated mode: a final bid at or above the reserve is awarded at closing. Brokerage mode: off, the seller always confirms
Seller decision window	72 h	No answer = not awarded
Counter-offer	On	Seller proposes one price; buyer has 24 h
Payment due	4 days after the invoice	Reminders after 2 and 3 days
Collection	Within 14 days of award	
Runner-up offer	48 h	Used when the winner defaults
ID check	Required for any bid $\geq$ €25,000	
Card hold for risky bidders	5% of the bid, at most €1,000 held per bidder	No completed purchase yet, or a past default
Premium basis	Start price of the lot	18% below €25,000 · 12% to €100,000 · 8% from €100,000
VAT on the premium	Rate of the platform's own country	France 20%, Netherlands 21%

Increments

Current bid from (€)	Next step (€)
0	50
1,000	100
5,000	250
10,000	500
25,000	1,000
50,000	2,000
100,000	5,000
250,000	10,000
1,000,000	25,000

This table reproduces every next-minimum bid observed on boatauction.com in October 2026 (€1,500 → €1,600, €10,500 → €11,000, €32,000 → €33,000, €68,000 → €70,000). The next minimum is the start price when there are no bids, else the current bid plus the step for its band.

Buyer total

$$\text{total} = H + \text{round}(H \cdot v_{\text{lot}}) + P + \text{round}(P \cdot v_{\text{fee}}), \quad P = \text{round}(H \cdot r(\text{start}))$$

H is the hammer price, r the premium rate for the lot's start price, v_{lot} the VAT rate on the boat (0 for private sellers and the margin scheme), v_{fee} the VAT rate on the premium. Every line rounds half-up to the cent.

Checks before the engine runs

The bidding service rejects the bid, with a specific error code, when any of these fail:

1. The bidder is signed in, has verified email and phone, and accepted the current terms.
2. The bidder is not the seller and not linked to the seller (same verified identity, phone, payment card, device or address).
3. The bidder is not suspended and has no unpaid invoice past due.
4. ID is verified when the amount is €25,000 or more; a card hold exists when the risk rules ask for one.

5. The bidder's total of leading bids stays within their credit limit (default €250,000, admin-editable).
6. Rate limit: at most 10 bid requests per minute per bidder per lot.
7. The amount is at most 20 times the next minimum; the interface asks for a second confirmation above twice the next minimum.

The engine

```
function applyBid(s: LotState, cmd: BidCommand, now: Instant, rules: Rules):
Result {
  if (s.status !== 'LIVE') return err('LOT_NOT_LIVE')
  if (now >= s.endsAt) return err('LOT_CLOSED')
  if (cmd.amount <= 0 || cmd.amount % 100 !== 0) return err('INVALID_AMOUNT')
  const next = s.price === null ? s.start : s.price + inc(s.price, rules)
  const reserveOpen = rules.reserveAware && s.reserve !== null && (s.price ??
0) < s.reserve
  const out: EmittedBid[] = []

  // A. The current leader raises their own max (a leader may never bid
against themselves)
  if (cmd.user === s.leader) {
    if (cmd.kind === 'single') return err('ALREADY_LEADING')
    if (cmd.amount <= s.leaderCeiling) return err('MAX_TOO_LOW')
    let price = s.price
    if (reserveOpen) price = cmd.amount >= s.reserve ? s.reserve : cmd.amount
    if (price !== s.price) out.push(bid(s.leader, price, 'auto'))
    return done(s, { leader: s.leader, ceiling: cmd.amount, ceilingAt: now,
price }, out, now, rules)
  }
  if (cmd.amount < next) return err(cmd.kind === 'single' ? 'BID_TOO_LOW' :
'MAX_TOO_LOW')

  // B. First bid on the lot
  if (s.leader === null) {
    let price = cmd.kind === 'single' ? cmd.amount : s.start
    if (cmd.kind === 'max' && reserveOpen) price = cmd.amount >= s.reserve ?
Math.max(s.start, s.reserve) : cmd.amount
    out.push(bid(cmd.user, price, cmd.kind === 'single' ? 'manual' : 'auto'))
    return done(s, { leader: cmd.user, ceiling: cmd.amount, ceilingAt: now,
price }, out, now, rules)
  }

  // C. Challenger against the leader's ceiling M (M = leader's max, or the
price if no max)
  const M = s.leaderCeiling, C = cmd.amount
```

```

    if (C > M) {
      if (M > s.price) out.push(bid(s.leader, M, 'auto')) // leader's
max is used up
      let price = cmd.kind === 'single' ? C : Math.min(C, M + inc(M, rules))
      if (cmd.kind === 'max' && reserveOpen) price = C >= s.reserve ?
Math.max(price, s.reserve) : C
      out.push(bid(cmd.user, price, cmd.kind === 'single' ? 'manual' : 'auto'))
      return done(s, { leader: cmd.user, ceiling: C, ceilingAt: now, price },
out, now, rules)
    }
    out.push(bid(cmd.user, C, cmd.kind === 'single' ? 'manual' : 'auto')) //
challenger bids in full
    const price = C === M ? M : Math.min(M, C + inc(C, rules)) //
tie: the earlier ceiling wins
    out.push(bid(s.leader, price, 'auto'))
    return done(s, { leader: s.leader, ceiling: M, ceilingAt:
s.leaderCeilingAt, price }, out, now, rules)
  }

function done(s, next, out, now, rules): Result {
  const visible = next.price !== s.price || next.leader !== s.leader
  const endsAt = visible && s.endsAt - now < rules.softWindowMs
    ? Math.max(s.endsAt, now + rules.extensionMs) : s.endsAt
  return ok({ ...s, ...next, endsAt, extensions: s.extensions + (endsAt >
s.endsAt ? 1 : 0) }, out)
}

```

Emitted bids get a per-lot sequence number and are stored append-only. The public history shows amounts, times and a stable pseudonym per bidder per auction ("Bidder 4821") with a country flag; it never marks which bids were automatic. A bidder sees "auto" only on their own bids.

Test vectors

Codex turns each row into a unit test. Amounts in euros; no reserve unless stated.

#	Lot before	Command	Bids emitted, in order	Price after	Leader after
1	Start 10,000, no bids	A max 14,000	A 10,000	10,000	A (ceiling 14,000)
2	After 1	B single 10,500	B 10,500 · A 11,000	11,000	A
3	After 2	B max 14,000	B 14,000 · A 14,000	14,000	A (earlier ceiling)
4	After 3	B max 15,000	B 14,500	14,500	B (ceiling 15,000)

#	Lot before	Command	Bids emitted, in order	Price after	Leader after
5	Start 150, A max 230	B single 220	B 220 · A 230	230	A (partial step, capped at max)
6	Start 5,000, A max 6,000	B single 7,000	A 6,000 · B 7,000	7,000	B
7	After 6	B single 7,500	none (ALREADY_LEADING)	7,000	B
8	After 6	B max 9,000	none, silent raise, no extension	7,000	B (ceiling 9,000)
9	Start 20,000, reserve 24,000	A max 30,000	A 24,000	24,000	A, reserve met
10	Start 20,000, reserve 24,000, A max 22,000 (bid in full)	B max 26,000	B 24,000	24,000	B, reserve met
11	Start 1,000, A max 2,000	B max 2,000	B 2,000 · A 2,000	2,000	A (tie, earlier)
12	Price 9,750	C single 9,900	none (BID_T00_LOW , minimum 10,000)	9,750	unchanged
13	Ends 20:00:00	Visible bid at 19:57:30	ends 20:02:30	—	—
14	Ends 20:00:00	Visible bid at 19:54:59	ends unchanged	—	—
15	Ends 20:00:00	Bid at 20:00:00.000	none (LOT_CLOSED)	—	—
16	Start 18,000, hammer 21,000, VAT on premium 21%	Invoice	Premium 3,780.00 · VAT 793.80	Total 25,573.80	—
17	Start 25,000, hammer 30,000, VAT 21%	Invoice	Premium 12% = 3,600.00 · VAT 756.00	Total 34,356.00	—

Invariants for property-based tests

Run thousands of random command sequences through the engine (fast-check) and assert after every step:

- Without voids, the price never goes down and is never below the start price.
- The leader's ceiling is at least the price; no other bidder holds a ceiling above the price.
- The winner never pays more than one step above the runner-up's ceiling, except after a single bid or a reserve jump.
- Equal ceilings: the earlier one leads.
- The closing time never moves earlier.

- Replaying the same commands with the same timestamps gives the same state, byte for byte.

How the server applies a bid

1. `POST /v1/lots/{id}/bids` with an `Idempotency-Key` header; a repeated key returns the first result.
2. Run the checks above, then open a transaction and `SELECT ... FOR UPDATE` the lot row.
3. Read the time from the database (`clock_timestamp()`), never from the client or the app server.
4. Call `applyBid`. Store the command, the emitted bids, the new lot state, an audit entry and outbox events, then commit.
5. After commit, the outbox relay publishes `lot.updated` to every viewer and queues outbid notifications. A changed closing time reschedules the close job.

Target: 95% of bids answered in under 150 ms on the server, with 50 bids per second on one lot during closing.

Closing on time

- Each live lot has one delayed close job at its `endsAt`, with an id built from the lot and that time, so duplicates collapse.
- The job locks the lot. If `endsAt` moved later, it schedules a new job and stops; otherwise it closes the lot and computes the outcome.
- A sweeper runs every 5 s and closes any live lot whose `endsAt` passed more than 2 s ago, in case a job was lost.
- A bid that reaches the database at or after `endsAt` is always rejected, even if the close job has not run yet.

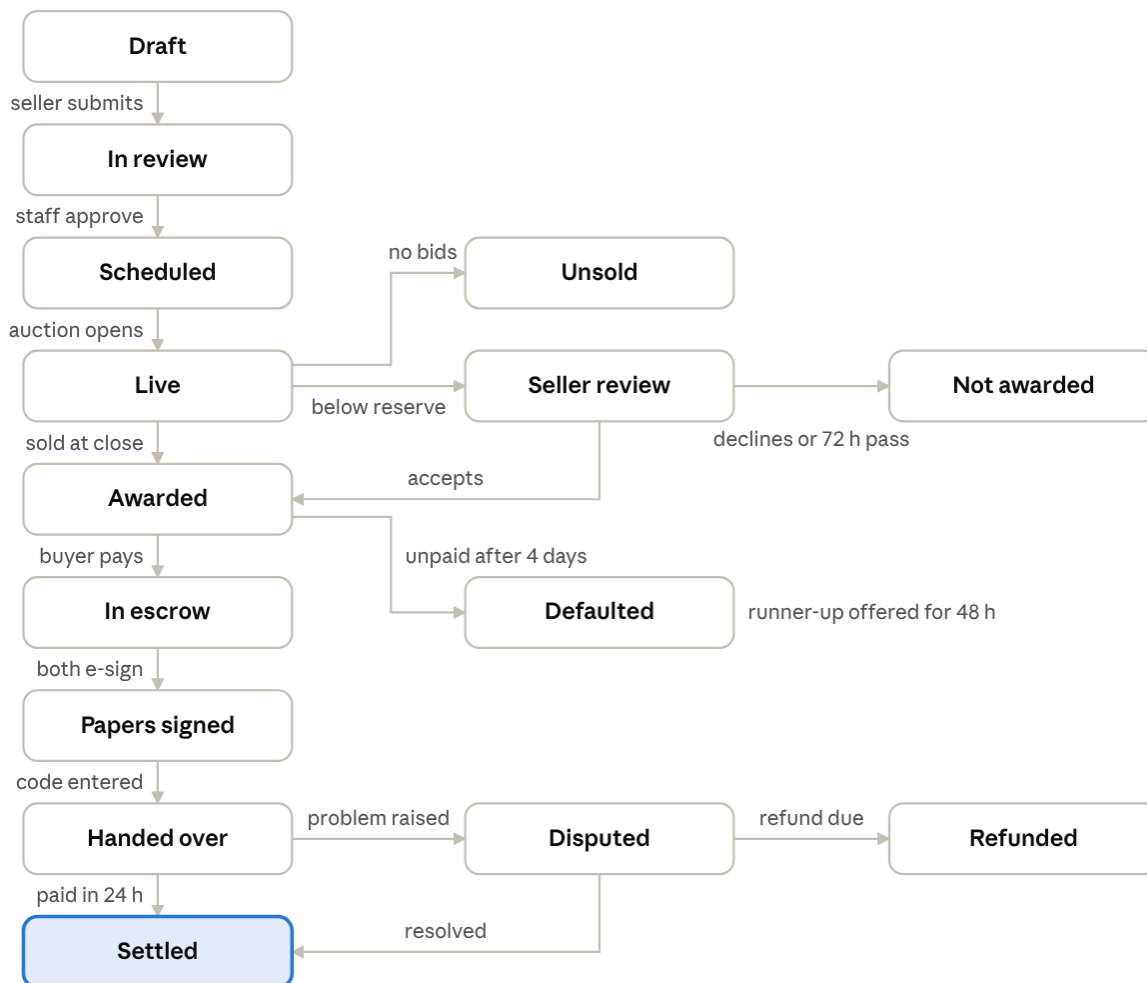
Outcome at closing

Situation at close	New state
No bids	Unsold, seller offered the next auction
No-reserve lot, regulated mode	Awarded at once
Reserve set and final bid $\geq$ reserve, regulated mode	Awarded at once
Every other case, and every lot with bids in brokerage mode	Awaiting seller, 72 h to accept, counter or decline

In brokerage mode the seller confirms every sale, so the no-reserve badge is unavailable and a reserve is only a target shown to the seller. A counter-offer must be higher than the top bid; if the buyer declines it or 24 h pass, the lot is not awarded. While a lot is live, its reserve can only be lowered, with the seller's written consent, and the engine then re-applies the reserve rule to the leader's ceiling. A reserve never goes up.

Lot lifecycle

A lot pays out only after escrow and a confirmed handover



lot lifecycle · 9 main states, 6 side states

The left column is the normal path. A lot without bids ends unsold, a declined or unanswered one ends not awarded, and a dispute decided for the buyer ends refunded; staff can also withdraw a lot before it closes. In brokerage mode, every lot with bids goes through seller review.

The same machine as a transition table, which is what Codex implements:

From	Event	To
Draft	Seller submits	In review
In review	Staff approve	Scheduled
In review	Staff reject, with reasons	Draft
Scheduled or Live	Staff withdraw (fee if bids)	Withdrawn
Scheduled	Auction opens	Live
Live	Closes with no bids	Unsold, relist offered
Live	Closes in regulated mode, no reserve or reserve met	Awarded
Live	Closes in any other case	Seller review
Seller review	Seller accepts, or buyer accepts the counter-offer	Awarded
Seller review	Seller declines, 72 h pass, or counter-offer refused or expired	Not awarded
Awarded	Buyer pays into escrow	In escrow
Awarded	Unpaid after 4 days	Defaulted
Defaulted	Runner-up accepts within 48 h	Awarded, runner-up as buyer
Defaulted	Runner-up declines or 48 h pass	Unsold, relist offered
In escrow	Both parties e-sign	Papers signed
Papers signed	Seller enters the buyer's code	Handed over
Handed over	24 h without a problem; 14 days for a consumer buying from a business outside public-auction mode	Settled, payout sent
In escrow, Papers signed or Handed over	Problem raised	Disputed
Disputed	Resolved for the seller	Settled, payout sent
Disputed	Resolved for the buyer, or a valid consumer withdrawal	Refunded

Voiding a bid

Bids are binding; bidders cannot withdraw them. Staff can void a bid for fraud or a proven system error, with a written reason and a second staff approval. The engine then replays the lot's remaining commands in their original order and timestamps, which recomputes price and leader; the closing time never moves earlier. Every affected bidder is notified, and the bid log keeps the voided entry, marked.

Seller acquisition: "Vendre mon bateau" and the cost-of-waiting estimator

Sellers are the scarce side of the marketplace, so every page pushes one action: "Vendre mon bateau". The hook is an honest number: what keeping the boat costs each month, computed from public cost data and the owner's own figures, with every assumption visible and editable.

Where the button lives

- Header on every page, in the accent colour, and a sticky bar on mobile.
- Homepage block: three inputs (type, length, value) give the monthly figure instantly, then "See the full breakdown".
- Every lot page: "Own a similar boat? See what it costs you to keep it."
- Results archive: "Sold for €X in this auction. Yours could be next."
- Footer, newsletter and the weekly "recently sold" email.

The funnel

1. **Hook.** The monthly cost appears after three answers, before any contact details.
2. **Refine.** Optional details sharpen it: country and coast, how the boat is kept, condition, real costs, days used per year.
3. **Result.** Monthly and yearly cost, breakdown chart, cost per day actually used, and "selling now instead of in 6 months saves about €X".
4. **Act.** "Sell in the next auction" opens the intake wizard pre-filled with every answer. Alternatives: "Email me this report" (PDF) or "Call me back".
5. **Follow-up.** Only with consent: a reminder 48 h after an unfinished intake, and the next auction's submission deadline.

The page stays honest, which is also the law: no fake countdowns, no pre-ticked consent boxes, no invented social proof. Urgency comes only from real facts, such as the actual submission deadline of the next auction.

The algorithm

The result is the yearly cost of keeping the boat, divided by 12:

$$C_{\text{year}} = B + I + M + W + T + D + O$$

Term	What it is	Default when the owner gives no figure	Basis
B	Berth or storage	France: $25 \times L^{2.25}$ € a year (L in metres), times a coast factor; the curve passes through the national averages	Observatoire national des ports de plaisance 2026 : about €1,400 for 6 m, over €21,000 above 20 m
I	Insurance	1% of value a year, at least €200	Banque Populaire : all-risk cover averages about 1% of value
M	Maintenance and repairs	The larger of 5% of value and €75 per metre	Professional rule of 5–10% of value a year, approximate
W	Winter storage ashore	€15 per metre per month, for the months entered (default 5) when the boat winters ashore	Port Adhoc 2026 tariff, Strijensas , approximate
T	Boat tax	France: the owner's TAEMUP amount, with a link to the official simulator ; the scale changes in 2027, so it is never hard-coded	mer.gouv.fr : hull of 7 m or more, due from €76
D	Loss of resale value	Value $\times$ yearly rate by age, where age = current year – year built: 0 $\rightarrow$ 15%, 1–5 $\rightarrow$ 7%, 6–10 $\rightarrow$ 5%, 11–20 $\rightarrow$ 4%, 21+ $\rightarrow$ 3%; $\times$ 0.85 excellent, 1.0 good, 1.25 fair, 1.5 needs work	Giornale della Vela : about 5% a year from year 2 to 5, flattening by year 10
O	Money tied up	Value $\times$ a safe savings rate, default 1.7%	Livret A at 1.7% since 1 August 2026

Coast factors for B start at 1.3 Mediterranean, 1.0 Corsica and national average, 0.85 Atlantic, 0.75 Channel and North Sea, 0.55 inland waterways. They are first estimates to calibrate with real seller data. Storage multiplies B: marina berth 1.0, dry stack 0.8, mooring

buoy 0.35, hard standing all year 0.5, trailer at home a flat €150. Netherlands, Belgium and Germany start at 0.8 of the French curve until local data replaces it.

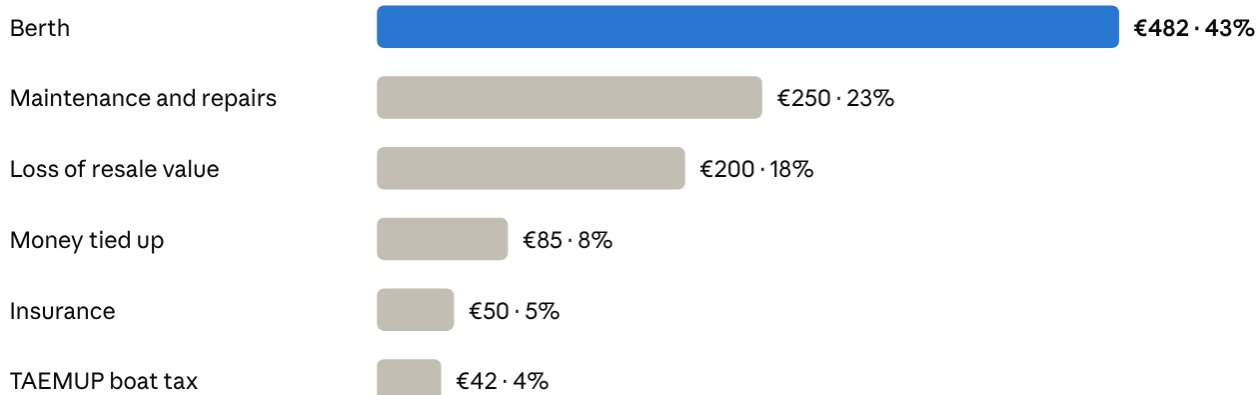
The owner's value V is required, with a slider and "Not sure? Our valuation is free". From the first results onward, the platform shows a median sale price per metre for similar boats from its own archive, never an invented figure.

Worked example

A 10 m motorboat built in 2008, worth €60,000, in good condition, on a Mediterranean berth all year, TAEMUP €500 from the owner's notice, used 15 days a year:

Keeping this boat costs about €1,108 a month; the berth is 43% of it

Monthly cost by item · €13,299 a year, €1,108 a month in total



10 m motorboat built 2008, worth €60,000, good condition, Mediterranean berth all year, used 15 days a year

worked example of this section · defaults from the algorithm table

Total €13,299 a year, so about €1,108 a month and €887 per day on the water. Selling in the next auction instead of in six months avoids about €6,650. Codex uses this case as a unit test: same inputs, €13,299 ± €1.

What the result screen shows

- Headline: "Your boat costs you about €1,108 a month."
- The breakdown chart, each line editable in place, with the source of every default one tap away.
- Cost per day used, when days are given.
- "Wait 3, 6 or 12 months" toggle with the cumulative figure.
- Selling-cost comparison: our flat seller fee against a broker commission of 8–10% ([Royal Nautisme Brokerage](#)), shown in euros for this boat.
- Next auction date and its real submission deadline.

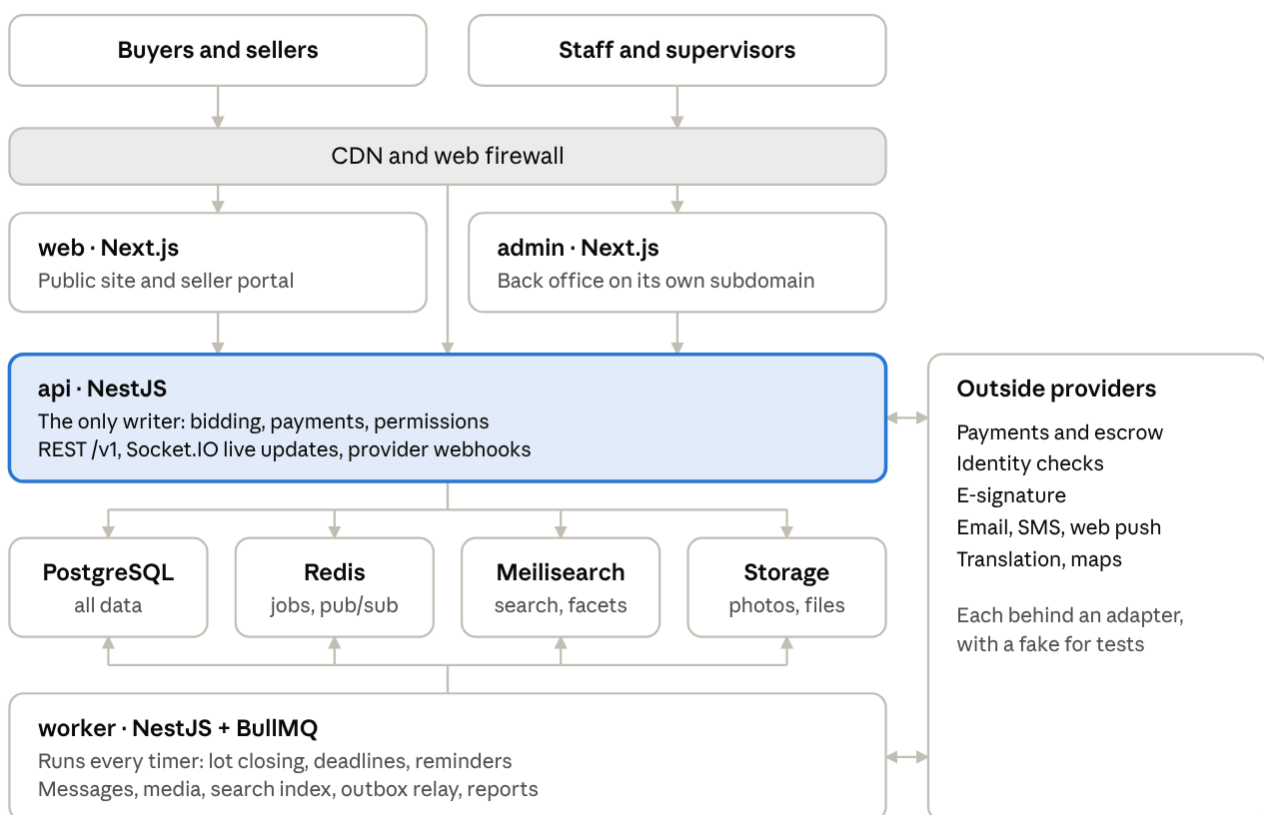
Built for change

- `estimateHoldingCost(input, params)` is a pure function in `packages/domain`.
- Every parameter lives in the admin panel; each change creates a new version, and each saved estimate stores the version it used.
- Each run is stored anonymously. A lead attaches to it only after consent. Admins see leads scored by value, region and stated timing.
- Version 2 replaces the default curves with medians from the platform's own results once each segment has at least 30 sales.

System architecture

Four deployable apps share one TypeScript monorepo: `web`, `admin`, `api` and `worker`. Only `api` writes business data, only `worker` runs timers, and both call the same pure rules in `packages/domain`.

Only the api writes business data; the worker runs every timer



system architecture · 4 apps, 4 data stores, outside providers

Browsers reach `web`, `admin` and the API's live updates through the CDN; `api` and `worker` share the same stores, and only they talk to outside providers.

Repository layout

```

apps/
  web/      Next.js: public site, buyer account, seller portal (fr, en, nl,
de)
  admin/    Next.js: admin panel on admin.<domain>
  api/      NestJS on Fastify: REST /v1, Socket.IO gateway, provider
webhooks
  worker/   NestJS standalone: BullMQ processors, schedulers, outbox relay
packages/
  domain/   pure rules: bidding engine, fees, increments, state machines,
estimator
  db/       Prisma schema, migrations, seed data, repositories
  contracts/ zod schemas shared by API and clients; OpenAPI generated from
them
  sdk/      typed API client generated from the OpenAPI file
  ui/       design system (Tailwind, Radix primitives), shared by web and
admin
  i18n/     messages for fr, en, nl, de; money, date and number formatters
  config/   eslint, tsconfig and prettier presets
  testing/  factories, fixtures, Testcontainers helpers
infra/     Terraform for AWS eu-west-3, Dockerfiles, k6 load tests
docs/     SPEC.md (this document) and architecture decision records

```

Components

Component	Owns	Scales by
<code>web</code>	Server-rendered pages, SEO, the estimator UI, the seller portal	More containers behind the CDN
<code>admin</code>	Every back-office screen; talks to <code>api</code> with staff sessions only	Small and fixed
<code>api</code>	Authentication, permissions, every write, bidding, Socket.IO, webhooks	More containers; Socket.IO shares rooms through Redis
<code>worker</code>	Lot closing, deadlines, notifications, media, search indexing, reports, the outbox relay	More containers per queue
PostgreSQL	All business data, audit log, outbox	Bigger instance, then read replicas for reporting

Component	Owns	Scales by
Redis	Job queues, pub/sub, rate limits, short caches	Managed cluster
Meilisearch	Catalogue search and facets, one index per locale	Rebuilt from PostgreSQL at any time
Object storage + CDN	Photos, videos, documents	Managed

Real-time

- One Socket.IO namespace `/live` with rooms `lot:{id}`, `auction:{id}` and `user:{id}`; the handshake checks the session.
- Every event carries `serverTime` and a per-lot `seq`. Clients keep a clock offset and redraw countdowns locally every 250 ms.
- A missing `seq` or a reconnect triggers `GET /v1/lots/{id}/state`, so a client never shows stale prices.
- If the socket fails, the page polls that endpoint every 10 s and shows "Live connection lost, prices refresh every 10 s".

Background jobs

Queue	Does
<code>lot-close</code>	One delayed job per live lot at its closing time
<code>deadlines</code>	Seller decision, counter-offer, payment and collection timers, plus reminders
<code>notifications</code>	Email, SMS, web push, in-app, per user preferences
<code>media</code>	Image sizes (AVIF, WebP), location metadata removed from photos, virus scan of documents
<code>search</code>	Index updates from outbox events; nightly full rebuild
<code>payments</code>	Webhook follow-up, bank-transfer matching, payouts
<code>reports</code>	DAC7 export, accounting exports, weekly seller reports, estimator PDFs

Outside providers, all behind adapters

Each provider is an interface in `packages/domain` with a real adapter and an in-memory fake for tests and local development: payments, identity checks, e-signature, email, SMS, web push, storage, search, translation, maps, analytics. Vendors are chosen in the Payments and Compliance sections.

Security

- **Sign-in.** Passkeys, email magic links, or password hashed with argon2id and checked against known breaches. Sessions in httpOnly, Secure, SameSite=Lax cookies, revocable per device.
- **Staff.** Passkey or company single sign-on is mandatory, sessions last 8 h, and sensitive actions ask for re-authentication.
- **Permissions.** Every API route declares one permission; ownership checks stop a seller from reading another seller's lots.
- **Web protections.** Strict content security policy, HSTS, rate limits in Redis, Cloudflare Turnstile on sign-up and contact forms, a WAF in front.
- **Sensitive data.** Identity documents sit in a separate bucket encrypted with KMS and open through 5-minute signed links; IBANs and tax numbers are encrypted in the application; logs redact personal data.
- **Supply chain.** Dependabot, CodeQL and secret scanning in CI; an external penetration test before launch.
- **Recovery.** Point-in-time database recovery for 35 days, daily snapshots copied to a second EU region, and a restore drill every quarter.

Observability

- OpenTelemetry traces across `web`, `api` and `worker`; JSON logs with a request id; Sentry for errors.
- Dashboards: bid latency, close-job lag, open sockets, queue depth, payment webhooks.
- Alerts: close-job lag above 2 s, bid p95 above 300 ms, error rate above 1%, any failed payout.

Environments and delivery

- **Local.** Docker Compose with PostgreSQL, Redis, Meilisearch, MinIO and Mailpit; `pnpm dev` starts everything with seed data.
- **Staging and production.** Same images. GitHub Actions builds, tests and pushes them; migrations run as a one-off task before each deploy, using expand-then-contract changes only.

- **Deploy freeze.** No production deploy from 17:00 to 23:00 on auction closing days.

Data model

PostgreSQL holds about 50 tables in six groups. Money is integer cents, time is UTC, bid and audit history is append-only, and secrets such as reserve prices sit in their own tables so a careless query cannot leak them.

Conventions

- **Ids.** UUIDv7 primary keys; human numbers (lot 7701, invoice 2026-000123) are separate columns.
- **Money.** `bigint` cents, EUR only in version 1; rates in basis points (1800 = 18.00%).
- **Time.** `timestampz` in UTC; displayed in the viewer's time zone, default Europe/Paris.
- **Concurrency.** A `version` column on every mutable row; bid writes lock the lot row.
- **History.** `bids`, `bid_commands`, `audit_log` and `ledger_entries` are insert-only; the app's database role has no UPDATE or DELETE on them. Voids and corrections are new rows.
- **Personal data.** Email as case-insensitive unique text, phone in E.164, tax numbers and IBANs encrypted in the application.

Identity and accounts

Table	Purpose	Key columns and rules
<code>users</code>	Every person, buyer or seller	email unique, phone, locale, country, status, risk level
<code>credentials</code> , <code>sessions</code>	Passkeys, password hash, magic links, device sessions	session revocable per device
<code>verifications</code>	Email and phone codes	expiry, attempts
<code>kyc_checks</code>	Identity checks and their results	provider, level, status, expires_at
<code>companies</code> , <code>company_members</code>	Business accounts	registration number, VAT number with VIES result, beneficial owners

Table	Purpose	Key columns and rules
<code>addresses</code> , <code>consents</code>	Postal addresses; terms and marketing consents	consent version and timestamp
<code>staff_roles</code> , <code>staff_permissions</code> , <code>staff_assignments</code>	Admin access control	see the Admin panel section

Catalogue

Table	Purpose	Key columns and rules
<code>categories</code>	Motorboat, sailboat, RIB, sloop, catamaran, jet ski, engine, trailer...	slug and name per locale
<code>attribute_definitions</code>	The spec template per category	key, type (number, text, enum, bool), unit, group, label per locale
<code>boats</code>	The vessel itself	brand, model, year, length, beam, draught (cm), hull material, CIN/HIN, registration country and number, VAT-paid status, CE category
<code>boat_attribute_values</code>	Values for the category template	one row per attribute
<code>media</code>	Photos and videos	storage key, sizes, order, alt text per locale
<code>documents</code>	Survey, papers, special conditions	visibility: public, registered bidders, staff only

Auctions and bidding

Table	Purpose	Key columns and rules
<code>auctions</code>	A dated sale	sale mode (brokerage or public auction), opens_at, closing_starts_at, interval_s, viewing day, settings version

Table	Purpose	Key columns and rules
lots	A boat in an auction	lot number, status, start price, no-reserve flag, price, leader, leader ceiling and its time, bid count, ends_at, original_ends_at, extensions, version
lot_private	Secrets of a lot	reserve price, seller notes; readable only by the seller and staff with lots.view_reserve
bid_commands	What a bidder asked for	kind, amount, idempotency key unique per user, database time, IP, device, accepted or rejection code
bids	Bids the engine emitted	lot, seq unique per lot, bidder, amount, manual or auto, command, hash of the previous bid + this one
bid_voids	Staff voids	bid, reason, requested_by, approved_by
max_bids	Private ceilings	one active row per lot and bidder; status active, exhausted or cancelled
bidder_aliases	Public pseudonyms	unique per auction
bidding_limits, card_holds	Credit limit and card holds per bidder	amount, provider reference, release date
watchlist, saved_searches	Follows and alerts	filters as JSON, alert frequency
viewing_registrations, lot_questions	Viewing-day sign-ups, buyer questions	answered_by, published flag

Sale, money and handover

Table	Purpose	Key columns and rules
award_decisions	Seller decision after closing	deadline, decision, counter amount and deadline, decided_by

Table	Purpose	Key columns and rules
<code>sales</code>	One per awarded lot	buyer, seller, hammer, premium, VAT amounts, status from awarded to settled
<code>second_chance_offers</code>	Offer to the runner-up	amount, deadline, answer
<code>invoices</code> , <code>invoice_lines</code>	Buyer invoices, seller fee invoices, credit notes	gapless number per series and year, PDF, e-invoicing status
<code>payments</code> , <code>refunds</code>	Money in and back	provider, method, virtual IBAN, provider reference, status
<code>ledger_accounts</code> , <code>ledger_entries</code>	Double-entry book of escrow funds	each transaction sums to zero; one escrow account per sale
<code>payouts</code>	Money to sellers	amount, IBAN snapshot, provider reference, status
<code>transfer_documents</code>	Bill of sale and transfer	template version, e-sign envelope, both signatures, PDF
<code>pickups</code>	Collection appointment	slot, place, hashed handover code, confirmed_at
<code>disputes</code>	Problems after the sale	type, status, outcome, funds frozen flag

Seller funnel

Table	Purpose	Key columns and rules
<code>estimator_runs</code>	Every estimate, anonymous until consent	inputs, parameter version, result, locale, campaign tags
<code>leads</code>	A seller prospect	run, contact, consent, score, pipeline stage, owner
<code>intake_drafts</code>	Unfinished intake wizard	step data, resume token, expiry
<code>consignments</code>	The seller's signed agreement	fees, exclusivity end date, e-sign envelope

Content and system

Table	Purpose	Key columns and rules
<code>pages</code> , <code>articles</code> , <code>faq_items</code> , <code>banners</code>	CMS content	one row per locale, SEO fields
<code>email_templates</code> , <code>ui_translations</code>	Message texts	key, locale, version
<code>settings</code>	Every business rule from this spec	key, JSON value, version, effective_from, changed_by
<code>audit_log</code>	Every staff and system change	actor, action, entity, before and after, IP, hash chain
<code>outbox_events</code> , <code>webhook_inbox</code>	Reliable events out and in	webhook event id unique per provider
<code>notifications</code>	Messages sent	channel, template, status
<code>feature_flags</code> , <code>dsa_notices</code> , <code>dac7_reports</code> , <code>gdpr_requests</code>	Operations and compliance	see the Compliance section

Database rules Codex must add

- `CHECK (amount_cents > 0 AND amount_cents % 100 = 0)` on bids and bid commands.
- `UNIQUE (lot_id, seq)` on bids; `UNIQUE (user_id, idempotency_key)` on bid commands; `UNIQUE (provider, event_id)` on the webhook inbox.

- Invoice numbers come from a counter row locked per series, so numbering has no gaps.
- A deferred trigger rejects any ledger transaction whose entries do not sum to zero.

API and real-time events

The API is JSON over HTTPS under `/v1`, described by an OpenAPI file generated from the zod schemas in `packages/contracts`. Bids and payments go through HTTP only; the socket only pushes updates out.

Conventions

- **Auth.** Session cookie for the web apps, with an Origin check on every write; bearer tokens reserved for future mobile apps.
- **Errors.** RFC 9457 problem JSON with a stable `code` (for example `BID_T00_LOW`) and `params` (for example `minimumCents`); the client translates the code.
- **Idempotency.** `Idempotency-Key` required on bids, payments, payouts and seller decisions; results kept 24 h.
- **Pages.** Cursor pagination, `?cursor=&limit=`, at most 100 items.
- **Money and time.** Integer cents plus currency; ISO 8601 UTC timestamps; every response carries `serverTime`.
- **Limits.** Per IP and per user, announced in `RateLimit` headers.

Endpoints

Method and path	Purpose	Access
GET <code>/auctions</code> , GET <code>/auctions/{slug}</code>	Calendar and auction pages	Public
GET <code>/lots</code> , GET <code>/lots/{id}</code>	Search with facets; lot detail	Public
GET <code>/lots/{id}/state</code>	Live state: price, leader alias, <code>ends_at</code> , bid count, seq	Public
GET <code>/lots/{id}/bids</code>	Public bid history	Public
POST <code>/estimator/runs</code>	Compute a cost-of-waiting estimate	Public, rate-limited
POST <code>/leads</code> , POST <code>/contact</code> , POST <code>/call-requests</code>	Seller leads and contact	Public, Turnstile

Method and path	Purpose	Access
POST /auth/*	Register, sign in, magic link, passkeys, sign out	Public
GET/PATCH /me, /me/bids, /me/watchlist, /me/saved-searches, /me/notification-settings	Buyer account	Signed in
POST /me/kyc/sessions	Start an identity check with the provider	Signed in
POST /lots/{id}/bids	Place a single or max bid	Bidder
DELETE /lots/{id}/max-bid	Stop automatic bidding; the latest bid stays binding	Bidder
POST /lots/{id}/viewing-registrations, POST /lots/{id}/questions	Viewing day, questions	Signed in
GET /me/sales/{id}	Won lot: invoice, payment, documents, pickup	Buyer
POST /sales/{id}/payments	Get the virtual IBAN or start a checkout	Buyer
POST /sales/{id}/counter-offer/accept, /reject	Answer a seller's counter-offer	Buyer
POST /sales/{id}/pickup, GET /sales/{id}/handover-code	Book collection; show the code	Buyer
POST /sales/{id}/handover	Seller enters the buyer's code	Seller
POST /sales/{id}/disputes	Report a problem	Buyer or seller
POST /seller/intakes, PATCH /seller/intakes/{id}, POST /seller/intakes/{id}/submit	Intake wizard with autosave	Signed in
POST /uploads	Presigned upload URL for photos and documents	Signed in
GET /seller/lots, GET /seller/lots/{id}/stats	Seller dashboard	Seller
POST /seller/lots/{id}/decision	Accept, decline or counter	Seller

Method and path	Purpose	Access
<code>/admin/*</code>	Every admin module; each route names its permission	Staff
<code>POST /webhooks/{provider}</code>	Payment, identity, e-sign, email events	Signature checked

Incoming webhooks are verified, stored in `webhook_inbox` and processed by the worker, so a repeated event never runs twice.

Bid rejection codes

`LOT_NOT_LIVE`, `LOT_CLOSED`, `AUCTION_PAUSED`, `INVALID_AMOUNT`, `BID_TOO_LOW`, `MAX_TOO_LOW`, `ALREADY_LEADING`, `AMOUNT_TOO_HIGH`, `TERMS_NOT_ACCEPTED`, `ID_CHECK_REQUIRED`, `CARD_HOLD_REQUIRED`, `LIMIT_EXCEEDED`, `SELLER_CANNOT_BID`, `ACCOUNT_SUSPENDED`, `RATE_LIMITED`. Each one has a translated message and, where useful, a button ("Verify my ID", "Raise my max bid").

Socket events

Event	Room	Payload	Sent when
<code>lot.updated</code>	<code>lot:{id}, auction: {id}</code>	lotId, seq, priceCents, leaderAlias, bidCount, endsAt, extended, status, reserveMet (if shown)	A bid changes the lot, or its status changes
<code>lot.bid</code>	<code>lot:{id}</code>	seq, alias, country, amountCents, at	Each emitted bid
<code>lot.closed</code>	<code>lot:{id}, auction: {id}</code>	outcome (sold, awaiting seller, unsold), final price if public	At closing
<code>user.outbid</code>	<code>user:{id}</code>	lotId, priceCents, nextMinimumCents	The user loses the lead
<code>user.leading</code>	<code>user:{id}</code>	lotId, priceCents	An automatic bid kept the user ahead
<code>user.notification</code>	<code>user:{id}</code>	id, type, title, body, link	Any in-app notification
<code>auction.updated</code>	<code>auction: {id}</code>	status, paused, message	Staff pause or resume an auction

Clients send only `subscribe` and `unsubscribe` with a list of lot ids, at most 200 per socket.

Admin panel

The admin panel controls every object and every rule on the platform, so no developer is needed for day-to-day operations after launch. Every action is checked against a permission and written to the audit log, and the risky ones need a second person.

Modules

Module	What staff can do
Dashboard	Live figures: lots live, bids per hour, total hammer value of the auction, sell-through rate, revenue, new leads, overdue payments, open disputes. On closing night, a live board of lots in closing order with clocks, extensions and viewers
Auctions	Create a sale (dates, first closing time, interval, viewing day, sale mode); order lots by drag and drop, which re-staggers closing times; publish; pause and resume all bidding (closing times shift by the pause length); extend one lot or all; cancel
Lots and listings	Review intakes; manage photos (order, crop, cover, blur faces and plates); edit descriptions with an AI draft and the four translations side by side; set start price, reserve, no-reserve and auto-award (regulated mode only), reserve badge, fee and VAT overrides, special conditions; feature on the homepage; withdraw with the fee computed; relist unsold lots
Bids and integrity	Live bid monitor; full bid log per lot, including private max bids; void a bid; a suspicious-activity queue (bidder shares a device, IP, phone, card or address with the seller; a new account bids only on one seller's lots; repeated small bids that only push the price up); block a bidder; set limits and card-hold rules
Users and companies	Search; profile, identity status and documents; manual verification; risk level and limits; suspend with a reason; notes and message history; GDPR export and erasure; read-only "view as user"
Sellers and leads	Lead inbox from the estimator, contact and call-back forms, with scores; a pipeline board (new, contacted, intake, photos, listed, sold or unsold, paid); owner, tasks and reminders; call log; consignment agreements; seller fee invoices

Module	What staff can do
Award desk	Closed lots awaiting the seller, sorted by deadline, with top bid, runner-up and reserve; record a decision taken by phone, with the call note; counter-offers; second-chance offers; defaults
Finance	Invoices and credit notes; match incoming bank transfers; escrow ledger per sale; payouts and refunds; late fees and break-up fees; exports (CSV, FEC for French accounting, UBL); VAT reports; e-invoicing status
Transfers and handover	Transfer documents and signature status; collection calendar; handover confirmations; disputes with frozen funds and evidence
Content and SEO	Pages, FAQ, guides, banners, spotlight, verified testimonials; SEO fields per page and locale; redirects
Communications	Email, SMS and push templates per locale with preview and test send; notification log with resend; newsletter export
Configuration	Every rule in the Auction rules section, versioned with an effective date; fee tables, increments, VAT rates, ID-check thresholds, estimator parameters, categories and their spec templates, auction calendar, feature flags, maintenance mode
Compliance	Identity and company review queue; sanctions and politically-exposed-person hits; DSA notices; DAC7 completeness and report files; GDPR requests; consent logs; audit log viewer; a read-only workspace for the bailiff or notary with signed bid-log exports
System	Queues and failed jobs with retry; webhook log; outbox lag; integration health; partner API keys for bulk listing; staff accounts and roles

Every list has saved views, bulk actions, CSV export and live updates. A global search finds lot numbers, emails, invoice numbers and the last four digits of an IBAN.

Roles and permissions

"Request" means the action waits for a second person with "Approve".

Capability	Super admin	Auction manager	Listing editor	Sales	Support	Finance	Compliance	Supervisor
Create, publish, pause auctions	✓	✓	view	view	view	view	view	view
Edit listings and content	✓	✓	✓	view	view	—	view	—
See and change reserve prices	✓	✓	—	view	—	—	view	—
Bid log with private max bids	✓	✓	—	—	—	—	✓	view
Void a bid	approve	request	—	—	—	—	approve	—
Suspend users, set limits	✓	—	—	—	request	—	✓	—
Identity documents	✓	—	—	—	—	—	✓	—
Leads and seller pipeline	✓	view	—	✓	view	—	—	—
Award desk	✓	✓	—	✓	view	view	—	view
Invoices and payments	✓	view	—	—	view	✓	view	—
Payouts and refunds	approve	—	—	—	—	request	—	—
Fees, rules and estimator settings	approve	request	—	—	—	request	—	—
Staff accounts and roles	✓	—	—	—	—	—	—	—
Audit log	✓	—	—	—	—	—	✓	bids only

Actions that always need two people

- Voiding a bid, cancelling a live auction, or editing a lot's price fields once it has bids.
- Changing fees, increments, deadlines or any rule that touches money.
- Any refund; any payout above €10,000; any payout to an IBAN changed in the last 7 days.
- Erasing a user's data.

The requester writes a reason, the approver sees the before and after, and the request expires after 24 h.

Audit

Every staff and system change records who, what, when, from which IP, the before and after values and the reason. Entries are chained by hash, cannot be edited, and can be exported for the bailiff, an accountant or a court.

Payments, escrow, invoicing and handover

The buyer's money never touches the platform's own bank account. It goes to a wallet at a licensed payment institution, stays there until the buyer confirms handover, and is then split automatically between seller and platform.

Provider

Default: [Mangopay](#), an e-money institution licensed in Luxembourg, built for marketplaces. It offers a wallet per user or per sale, a [dedicated virtual IBAN per wallet, escrow, split payments, payouts and built-in KYC/KYB](#), and already runs Vinted, Wallapop and ManoMano. Lemonway and Stripe Connect are the alternatives to quote against it; the `PaymentProvider` adapter keeps the switch cheap.

Why not hold the money ourselves: holding buyers' funds for others is a regulated payment service in the EU. boatauction.com uses a Dutch third-party funds foundation; a licensed provider is the simpler and safer equivalent for a new company.

From award to payout

1. **Invoice.** Within 5 minutes of the award the buyer gets a payment statement: hammer price, VAT on the boat if the seller is a business, buyer's premium and VAT on the premium.
2. **Pay within 4 days.** Bank transfer to a virtual IBAN unique to this sale, so it matches automatically; SEPA Instant is recommended. Cards are accepted only up to €5,000, because card fees on a boat price are too high.
3. **Escrow.** The full amount stays in the sale's wallet. The buyer sees "Paid, held in escrow"; the seller sees "Buyer has paid".
4. **Transfer document.** Generated from a template with both identities, the boat's identifiers (CIN or HIN, registration, engine serials), its VAT status and the price, then e-signed by both parties.
5. **Collection.** The buyer books a slot within 14 days of the award. The app shows the buyer a 6-digit handover code and QR code.

6. **Handover.** The seller enters the buyer's code in the app at the boat. That single step proves the handover, with time and place.

7. **Release.** Within 24 hours (after 14 days in the consumer case below): hammer price minus any unpaid seller fee to the seller's verified IBAN, premium and fees to the platform's wallet. Both get a statement.

The buyer becomes owner after paying and signing, or 14 days after award at the latest, matching the competitor's terms; the app reminds them to insure the boat from that moment. A country checklist from the CMS explains the registration change, for example the French francisation or a Dutch notarial deed for a registered ship.

When something goes wrong

Situation	What the system does
Buyer has not paid after 2 and 3 days	Reminder by email, SMS and app
Buyer has not paid after 4 days	Sale cancelled; break-up fee invoice (premium + VAT); any card hold captured; bidding suspended; runner-up offered the lot at their last bid for 48 h, else relisted
Seller refuses handover or the boat is materially not as described	Dispute opened, funds frozen; if upheld, full refund to the buyer and a fee to the seller (premium + VAT on the price, at least €500)
Problem reported within 24 h of handover	Release paused until staff close the dispute

Boats are sold as seen. Disputes cover non-delivery, a different boat, missing papers or undisclosed debts on the boat, not cosmetic opinions.

One exception to "sold as seen": when a business seller sells to a consumer outside a public-auction mode, the consumer may cancel within 14 days of receiving the boat. For those sales the payout waits until the 14 days have passed, and a withdrawal request ends in Refunded. Who bears the fees and the return of the boat is decided by the lawyer and stored in settings.

Invoices

- **Who invoices what.** The platform invoices its premium to the buyer and its fee to the seller. A business seller's boat invoice is issued by the platform on the seller's behalf under a billing mandate in the consignment agreement. A private seller issues no invoice; the transfer document is the bill of sale.

- **Numbers.** Gapless per series and year: `B-2026-000123` buyer, `S-2026-000045` seller, `C-2026-000007` credit notes.
- **Content.** Legal names, addresses, company and VAT numbers, date, number, description, amounts and VAT per rate, payment terms; for business customers in France, the late-payment penalty and the €40 recovery fee.
- **Exports.** CSV and FEC for the accountant, plus e-invoicing through a certified platform where the law requires it (see Compliance).

An accountant should validate this invoicing model, and the VAT treatment of the premium for each country, before launch.

Escrow ledger

Every movement is a balanced double entry across `buyer_receivable`, `escrow:{sale}`, `seller_payable`, `platform_revenue`, `vat_collected` and `refunds`. A nightly job compares the ledger with the provider's wallet balances and alerts on any difference of one cent or more.

Compliance and legal

The French auction regime is the decision that shapes the product most, and a French lawyer must settle it before launch. The platform supports both legal modes per auction, so the answer changes settings, not code. This section is a builder's checklist, not legal advice.

The French auction regime

Under [Article L321-3 of the Code de commerce](#), offering a good online as the owner's agent and awarding it to the highest bidder is a regulated electronic auction sale.

"Courtage aux enchères" is outside that regime only if there is no award to the highest bidder and no third party involved in the description of the good and the conclusion of the sale. A courtage platform must also tell the public clearly what kind of service it is.

Our model mixes the two: staff improve descriptions, while the seller decides after closing. So each auction carries a `sale_mode` :

Mode	How it works	What changes in the product
brokerage	The seller writes and approves the listing; staff only check it; the seller accepts or declines the top bid	Banner "This is not a public auction" on every lot; descriptions marked as written by the seller

Mode	How it works	What changes in the product
regulated	A licensed auction house or commissaire de justice runs the sale and awards the lot to the highest bid at or above the reserve	Partner named on lots, terms and invoices; staff may write descriptions; award is automatic at closing

The Dutch model of boatauction.com (bailiff supervision, bidders may attend the closing in person) is a third variant for the Netherlands. The supervisor role, attendance sign-up and signed bid-log exports already support it.

Checklist

Topic	Rule (as of October 2026)	What the platform does
Withdrawal right	EU consumer law gives 14 days to cancel a distance purchase from a business, except at public auctions consumers may attend; online-only auctions do not count	Business seller selling to a consumer outside a public-auction mode: the lot shows the withdrawal right and the flow supports it. Private-to-private sales are not covered by that law
DSA notices	Every hosting service needs a notice-and-action channel	Report button on lots and questions; a DSA queue in admin with decisions and reasons sent to both sides
DSA trader traceability	Articles 30–32 (trader data, compliance by design, informing buyers) do not apply to micro or small platforms, or for 12 months after outgrowing that status (Article 29)	Built anyway for business sellers: name, address, contact, ID, payment account, trade register number and self-certification before a listing goes live
DAC7	Platforms report seller data yearly by 31 January; only sellers with fewer than 30 sales and €2,000 or less are excluded; after two reminders and 60 days, the platform must withhold payment or close the account (Taylor Wessing)	Almost every boat seller is reportable: tax number, date of birth, address and IBAN collected at onboarding; payout blocked until complete; yearly XML report and a copy for each seller

Topic	Rule (as of October 2026)	What the platform does
Anti-money-laundering	From 10 July 2027 the EU caps cash payments at €10,000 and treats sellers of watercraft above €7.5 million as obliged entities (CMS)	No cash, ever; every party verified through the payment provider; sanctions screening; lots above €7.5 million blocked until a compliance programme exists
E-invoicing in France	Every company must receive e-invoices from 1 September 2026; SMEs must issue them from 1 September 2027, through an approved platform (Weblex)	Invoices generated as Factur-X or UBL and sent through an approved platform's API; ready before September 2027
VAT	Premium taxed at the platform's country rate; the boat's VAT depends on the seller (private: none; dealer: margin scheme or standard rate); EU business buyers with a valid VAT number may get reverse charge	VAT regime per lot; VIES check of VAT numbers; boats under 3 months old or 100 engine hours flagged for staff review, since new-boat VAT rules differ; accountant sign-off
GDPR and cookies	Refusing cookies must be as easy as accepting; data minimised and kept no longer than needed	Consent banner with equal buttons and logs; EU hosting; data processing agreements with every provider; impact assessment for identity checks and fraud scoring; retention table in settings
Accessibility	The European Accessibility Act covers consumer e-commerce since June 2025, with an exemption for micro-enterprises	WCAG 2.2 AA from day one, checked in CI and by a manual audit before launch
Boat papers	Buyers need ownership, registration, CE and VAT-paid proof	Seller uploads them at intake; staff verify; lots show which documents exist; transfer checklist per country

Documents a lawyer drafts before launch

- ☐ General terms for buyers, one version per sale mode
- ☐ Consignment agreement for sellers, with exclusivity, withdrawal fee and billing mandate

- ☐ Privacy and cookie policies; legal notice; DSA contact point
- ☐ Auction rules page, written from the Auction rules section of this spec
- ☐ Templates: transfer document, special conditions, payment statement, invoices

Quality, launch and auction night

The bidding engine and every money path get the strictest tests, and launch waits for a full rehearsal auction with real people and real payments.

Tests

Layer	Tool	Covers	Gate in CI
Domain unit	Vitest + fast-check	Engine test vectors and invariants, fees, increments, state machines, estimator	100% branch coverage of packages/domain
Integration	Vitest + Testcontainers	Bid transaction, idempotency, outbox, close job, webhooks, ledger balance	All pass
Concurrency	Vitest	200 parallel bids on one lot: final state equals a one-by-one replay, no duplicate seq, one leader	All pass
API contract	OpenAPI diff	No breaking change without a new version	No breaking diff
End to end	Playwright	Both journeys, plus voiding a bid with approval, pausing an auction, the award desk; 4 locales; mobile size	All pass on staging
Load	k6	5,000 open sockets; 50 bids/s on one lot for 10 min	Bid p95 under 150 ms, close lag under 1 s
Failure	Scripted	Worker killed during closing; one API instance killed	Lots close within 7 s; clients resync

Layer	Tool	Covers	Gate in CI
Security	ZAP baseline, CodeQL	Common web flaws, unsafe code	No high findings
Accessibility	axe in Playwright	WCAG 2.2 AA rules	No serious violations

Performance budgets

- Lot page: largest contentful paint under 2.5 s on mid-range mobile (75th percentile), interaction delay under 200 ms, layout shift under 0.1, under 200 KB of compressed JavaScript.
- API: reads under 200 ms and bids under 150 ms at the 95th percentile.
- Live updates reach viewers within 300 ms of the bid being committed.

Before the first public auction

- ☐ Legal mode chosen and terms signed off by a lawyer
- ☐ Invoicing and VAT approved by an accountant
- ☐ Payment provider live; identity checks tested with real documents
- ☐ External penetration test done; high findings fixed
- ☐ Database restore drill done
- ☐ Load test passed at twice the expected peak
- ☐ Rehearsal auction: 20 internal bidders, real low-value lots, payment, signature and handover end to end
- ☐ Native speakers reviewed fr, en, nl and de
- ☐ Sitemaps, structured data and redirects checked
- ☐ On-call rota and a public status page for the first three auctions

Auction-night runbook

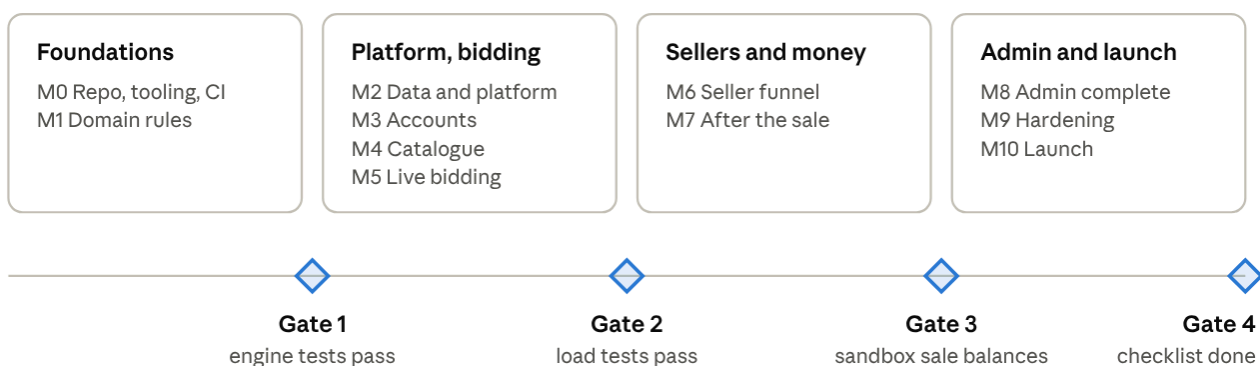
1. **Day before.** Deploy freeze on. Check every lot's closing time, the payment provider's status page and that reminders went out.
2. **Two hours before.** On-call engineer and auction manager online; dashboards open; server clocks synced; database load under 50%.
3. **Thirty minutes before.** Live closing board on screen; support chat staffed.
4. **During closing.** Watch close-job lag, bid latency and open sockets. An outage at the identity or payment provider never blocks bidders who are already verified.

5. **If bidding is impaired for more than 60 s.** Pause the auction with the kill switch: closing times shift by the pause length, a banner and an email tell watchers and bidders, and bidding resumes once stable. Write up the incident the same night.
6. **After the last lot.** Confirm every lot closed, the award desk queue is filled, invoices went out, and the post-auction report reached the team.

Build plan for Codex

Eleven milestones, each a series of small pull requests, with four gates that must pass before the next phase starts. The pure domain package comes first because bidding, money and the estimator all depend on it.

Each phase starts only after the gate before it passes



build roadmap · 11 milestones in 4 phases, 4 gates, not to scale

Gate 1 keeps app code off untested rules; Gate 3 lets real money flow only after a sandbox sale balances to the cent.

Milestones

Milestone	Tasks, one pull request each	Done when
M0 Foundations	Monorepo with pnpm and Turborepo; strict TypeScript, ESLint, Prettier; Docker Compose; CI; empty web , admin , api , worker with health checks; i18n and design-token scaffolding; AGENTS.md files	pnpm dev starts all four apps; CI green on an empty PR

Milestone	Tasks, one pull request each	Done when
M1 Domain core	Money and time helpers; increments and fees; <code>applyBid</code> and <code>done</code> ; lot and sale state machines; deadline calculator; estimator	Every test vector and invariant in this spec passes; 100% branch coverage. Gate 1
M2 Data and platform	Prisma schema and migrations; seed data; settings service with versions; audit log; outbox and relay; staff roles skeleton	Migrations run from zero; seed creates a demo auction; audit entries written by a sample action
M3 Accounts	Sign-up and sign-in (passkeys, magic link, password); email and phone verification; profile, consents; identity-check adapter with fake and real provider; staff single sign-on	A user can register, verify and reach the bidding gate; staff need a passkey
M4 Catalogue	Categories and spec templates; media pipeline; auctions and lots in admin; search index; catalogue and lot pages server-rendered in 4 locales; watchlist	A lot created in admin is searchable and indexable in all locales
M5 Live bidding	Bid endpoint and transaction; Socket.IO; close jobs and sweeper; outbid and closing-soon notifications; bid history; My bids; admin bid monitor; pause switch; void with two approvals	Concurrency, load and failure tests pass. Gate 2
M6 Seller funnel	"Vendre mon bateau" pages; estimator API, result page and PDF; leads pipeline; intake wizard; seller verification and DAC7 data; consignment e-sign; seller dashboard	An estimate turns into a signed consignment without staff typing anything

Milestone	Tasks, one pull request each	Done when
M7 After the sale	Award desk; seller decisions and counter-offers; invoices; payment provider (virtual IBAN, escrow, payouts); ledger; transfer e-sign; pickup and handover code; defaults, runner-up offers, disputes	A sandbox sale runs from award to payout with a balanced ledger. Gate 3
M8 Admin completeness	Dashboards; finance exports; CMS; message templates; every settings screen; compliance queues; audit viewer; supervisor workspace	Every rule in this spec is editable in admin, and every edit is audited
M9 Hardening	Security headers and limits; accessibility fixes; performance budgets; dashboards and alerts; restore drill; runbooks	All Quality gates green; penetration-test findings closed
M10 Launch	Legal sign-off; live payment account; rehearsal auction; first public auction	Launch checklist complete. Gate 4

Build each domain's admin screens in the same milestone as the domain, so nothing ships without its controls.

Keep Codex on track

- Codex reads `AGENTS.md` before any work and gives precedence to the file closest to the code it edits ([OpenAI docs](#)). Keep the root file short, and add `packages/domain/AGENTS.md` and `apps/api/AGENTS.md` with the local rules.
- This spec is larger than the instruction budget Codex loads automatically, so every task prompt names the spec sections to read.
- One task per conversation and per pull request. Ask for the plan first, approve it, then let it build.
- Never let Codex change `packages/domain` and an app in the same pull request, except in M1.

Prompt template

Read AGENTS.md, then these sections of docs/SPEC.md: <sections>.
 Task: <one task from the milestone table>.
 Stay inside: <folders>. Do not change anything else. Add no dependency that the spec does not name.
 First reply with a plan only: files to create or change, tests to add, open questions. Wait for my approval.
 Then implement, run `pnpm lint && pnpm typecheck && pnpm test`, and open a pull request with a summary, test output, screenshots for any UI, and every assumption you made.
 Done when: <the milestone's acceptance criteria>.

First three prompts, ready to paste

M0.1 — Read AGENTS.md and docs/SPEC.md sections "Start here" and "System architecture".
 Create the monorepo exactly as in "Repository layout": `pnpm workspaces`, Turborepo, strict TypeScript, ESLint and Prettier presets in packages/config, Vitest in every package, Playwright in apps/web.
 Add docker-compose.yml with PostgreSQL 17, Redis 7, Meilisearch, MinIO and Mailpit.
 Create empty apps web (Next.js), admin (Next.js), api (NestJS on Fastify), worker (NestJS standalone), each with a /health endpoint, and a GitHub Actions workflow running lint, typecheck, test and build.
 Plan first. Done when: `pnpm dev` starts all four apps and CI passes.

M1.2 — Read AGENTS.md and docs/SPEC.md section "Auction rules and bidding engine" in full.
 Implement packages/domain/src/auction: `types`, `inc()`, `nextMinimum()`, `applyBid()` and `done()` exactly as specified, plus `replay(commands)`. No I/O, no `Date.now()`: time is an argument.
 Turn every row of the "Test vectors" table into a Vitest test, and every "Invariants" bullet into a fast-check property with at least 10,000 runs. Plan first.
 Done when: all tests pass and branch coverage of packages/domain/src/auction is 100%.

M5.1 – Read AGENTS.md, apps/api/AGENTS.md and docs/SPEC.md sections "Auction rules and bidding engine", "Data model" and "API and real-time events".

Implement POST /v1/lots/{id}/bids in apps/api: the checks before the engine, Idempotency-Key handling, one transaction with SELECT ... FOR UPDATE on the lot, database time via clock_timestamp(), applyBid from packages/domain, inserts into bid_commands, bids, max_bids, audit_log and outbox_events, then commit.

Add an integration test with Testcontainers that fires 200 concurrent bids on one lot and compares the final state with a sequential replay. Plan first.

Done when: the concurrency test passes 20 runs in a row.

AGENTS.md

Copy the first block to AGENTS.md at the repository root, and the two short blocks to the folders they name. The root file stays well under Codex's default instruction budget of 32 KiB.

```
# AGENTS.md – Tidebid boat auction platform

## What this is
Timed online auctions for boats, in fr, en, nl and de. The spec is
docs/SPEC.md and it is the single
source of truth. Read the sections your task names before coding. If the spec
is unclear or
contradicts itself, stop and ask in the pull request. Never invent a business
rule.

## Commands
- Install: pnpm install
- Services: docker compose up -d    (PostgreSQL, Redis, Meilisearch, MinIO,
Mailpit)
- Run everything: pnpm dev
- Checks before every commit: pnpm lint && pnpm typecheck && pnpm test
- One package: pnpm --filter @tidebid/domain test
- End to end: pnpm e2e
- Database (local only): pnpm db:migrate, pnpm db:seed, pnpm db:reset
- Use pnpm only, never npm or yarn.

## Repository map
apps/web, apps/admin, apps/api, apps/worker
packages/domain (pure rules), db, contracts, sdk, ui, i18n, config, testing
```

infra/ (Terraform, Docker, k6), docs/ (SPEC.md, decision records)

Golden rules

1. Business rules live only in packages/domain as pure functions: no I/O, no `Date.now()`, no randomness.
Time and settings are arguments.
2. The server decides price, leader, closing time and outcome. Clients only display them.
3. Every bid goes through `BiddingService.placeBid()`: one transaction, `SELECT ... FOR UPDATE` on the lot,
database time from `clock_timestamp()`, then `applyBid()`. Nothing else writes a lot's price, leader or `ends_at`.
4. Money is integer cents (`bigint` in the database). Rates are basis points. Never floats for money.
5. Timestamps are `timestamptz` in UTC. Format them only at the edge, with `packages/i18n`.
6. `bids`, `bid_commands`, `audit_log` and `ledger_entries` are insert-only. Corrections are new rows.
7. Every change to a lot, bid, sale, payment or payout writes an `audit_log` row and an outbox event
in the same transaction.
8. Never expose reserve prices, max bids or bidder identities in public APIs or logs. Bidders appear as aliases.
9. Every API route declares a permission. Admin routes need a staff session with passkey or single sign-on.
10. Endpoints that bid or move money require an `Idempotency-Key`.
11. No user-facing text in code: add keys to `packages/i18n` for `fr`, `en`, `nl` and `de`.
12. Migrations are additive: expand now, contract in a later pull request. Nothing destructive without
an explicit instruction.
13. Never log personal data, tokens, IBANs or identity documents. Use the logger's redaction list.
14. Fees, increments, deadlines and thresholds come from the settings service, never from constants.

Code style

- TypeScript strict, no any. zod validates every external input.
- Errors are RFC 9457 problem objects with a stable code.
- Named exports; default exports only where Next.js requires them.
- Tests sit next to the code as `*.test.ts`. Domain tests use the spec's test vectors verbatim.
- UI uses `packages/ui` and Tailwind. Every control has a label, a visible focus state and AA contrast.

Definition of done for every pull request

- [] lint, typecheck, unit and integration tests pass; end-to-end tests too when UI changed
- [] new behaviour has tests; packages/domain keeps 100% branch coverage
- [] i18n keys exist in all four locales
- [] any new business object has its admin screen and audit entries
- [] docs/SPEC.md updated when behaviour changed, or the question raised in the pull request
- [] description lists the summary, test output, UI screenshots and every assumption

Never

- Add a dependency the spec does not name without asking.
- Change packages/domain and an app in the same pull request, except during milestone M1.
- Deploy, or change infra/, unless the task says so.
- Commit secrets. Use .env.example placeholders.
- Copy logic from docs/prototypes/. Those files are visual references; docs/SPEC.md defines the logic.

packages/domain/AGENTS.md

- Pure functions only: inputs in, outputs out. Never import from apps/* or packages/db.
- Every exported function has unit tests. Engine invariants are fast-check properties with at least 10,000 runs.
- Spec sections: "Auction rules and bidding engine", "Seller acquisition".

apps/api/AGENTS.md

- Bids only through BiddingService: transaction, row lock on the lot, clock_timestamp().
- Publish to sockets only after commit, through the outbox relay; never inside the transaction.
- Webhooks: verify the signature, store in webhook_inbox, let the worker process them.